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Laser-assisted pontic site soft tissue healing prior to fixed partial dentures: a randomized placebo-controlled clinical trial

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Abstract

Background This randomized, placebo-controlled clinical trial evaluated the clinical and biological effects of laser-assisted therapy on the healing of soft tissue at the pontic site prior to fixed partial denture rehabilitation.

Methods A total of 45 patients requiring a single-tooth extraction, followed by a three-unit fixed partial denture rehabilitation, were randomly allocated to one of three groups ($n=15$ in each group). Gallium–Aluminum–Arsenide (GaAlAs) diode laser (GRR laser), Neodymium-doped Yttrium Aluminium Garnet laser (Nd: YAG) and placebo. Laser applications were performed once daily for five consecutive days following tooth extraction. Clinical healing was assessed using wound dimension measurements and the healing index (HI) at baseline, day 7, and day 14. Postoperative pain was evaluated using visual analog scale (VAS) scores recorded daily from postoperative day 1 to day 7. Gingival crevicular fluid (GCF) samples were collected at the same time points and analysed for transforming growth factor beta 2 (TGF-β2) levels using an enzyme-linked immunosorbent assay. Intergroup and intragroup comparisons were performed using the appropriate statistical tests ($\alpha=0.05$).

Results No significant differences were observed among groups at baseline for any evaluated parameter ($p>0.05$). At day 7 and day 14, significant intergroup differences were detected for wound dimensions, clinical healing index scores, VAS pain scores, and GCF TGF-β2 levels ($p<0.001$). The GRR laser group demonstrated significantly greater wound size reduction, higher clinical healing index scores, lower postoperative pain levels, and higher TGF-β2 concentrations compared with the Nd: YAG laser and placebo groups. Nd: YAG laser group showed improved outcomes compared with placebo, although these differences were less significant than those observed in the GRR group.

Conclusions Laser-assisted therapy accelerated the healing of soft tissue at the pontic site prior to fixed partial denture rehabilitation. The GRR laser protocol achieved superior clinical and biological outcomes compared to the Nd: YAG laser and the placebo. These results suggest that optimised low-level laser therapy (LLLT) could be a valuable additional treatment for accelerating pontic site maturation and making the prosthetic workflow more predictable.

Trial registration NCT07415434, Retrospectively registered (09.02.2026).

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Keywords Clinical Trial, Fixed Partial Denture, Pontic, Soft Tissue Healing, Low-Level Laser Therapy

Background

Fixed partial dentures (FPDs) are a widely used and predictable treatment option for the replacement of single or multiple missing teeth, particularly in cases where implant therapy is contraindicated or not preferred by the patient [1]. The long-term clinical success of FPDs is influenced by various factors, including the mechanical integrity of the prosthesis and abutment teeth, as well as the quality, stability, and health of the surrounding soft tissues, particularly in the pontic region [2]. Pontic design plays a critical role in achieving optimal biological outcomes across all regions of the dentition, as well as esthetic outcomes particularly in the anterior and esthetically demanding areas in fixed prosthodontics. Various pontic designs—such as ridge-lap, modified ridge-lap, hygienic, and ovate pontics—have been described, each with specific indications based on anatomical, functional, and esthetic requirements [3]. Pontic design directly influences the biological response of the underlying soft tissues by affecting pressure distribution, plaque accumulation, tissue oxygenation, and the maintenance of epithelial integrity in the pontic region [4]. Therefore, rapid and predictable soft tissue healing at the pontic site is essential for successful prosthetic rehabilitation.

Delayed or inadequate soft tissue healing in the pontic area may lead to several prosthetic complications, including mucosal inflammation, ulceration beneath the pontic, patient discomfort during function, compromised oral hygiene, and difficulties in obtaining accurate impressions for definitive prosthetic fabrication [5]. Furthermore, if the soft tissues at the pontic site are not sufficiently mature, this may necessitate prolonged provisionalisation, repeated adjustments to the pontic, or a delayed transition to the definitive prosthesis. This may prolong the overall treatment duration, increase postoperative pain levels, and delay prosthetic rehabilitation [6]. From a biological perspective, rapid and uncomplicated healing of the pontic site is essential for achieving a stable mucosal seal, maintaining tissue contours, and ensuring long-term soft tissue compatibility. Soft tissue maturation is particularly important in pontic designs that apply controlled pressure to the mucosa. Excessive or prolonged inflammation can damage the integrity of the epithelium and disrupt the organisation of the connective tissue [5]. Therefore, clinical approaches that promote accelerated and high-quality soft tissue healing in the pontic region are of significant interest in prosthodontic practice.

Laser-assisted therapies have been increasingly investigated as a complementary treatment for enhancing soft tissue wound healing in dentistry. LLLT has been shown to accelerate wound healing by modulating cellular metabolism and enhancing mitochondrial activity. It also promotes the early resolution of inflammation without causing thermal tissue damage [7–9]. It has been reported that low-level laser applications stimulate fibroblast proliferation, collagen synthesis, angiogenesis and the release of growth factors involved in tissue regeneration [10–12]. These biostimulatory effects are particularly important for oral soft tissues, where rapid epithelialisation and connective tissue maturation are essential for achieving optimal function and prosthetic outcomes. Transforming growth factor beta (TGF-β) is one of the growth factors that play a central role in regulating the production of the extracellular matrix and the remodelling of connective tissue during wound healing [13]. TGF-β comprises three major isoforms (TGF-β1, TGF-β2, and TGF-β3), each isoform exhibits distinct biological functions during wound healing and tissue remodeling. TGF-β1 is the most extensively studied isoform and is primarily associated with early inflammatory responses and fibrosis. In contrast, TGF-β2 has been shown to play a critical role in epithelial–mesenchymal interactions, extracellular matrix organization, and soft tissue maturation. Importantly, TGF-β2 plays a role in regulating collagen deposition and tissue formation, which are essential for achieving stable and esthetically acceptable soft tissue contours. The assessment of soft tissue healing quality in relation to prosthetic rehabilitation requires not only clinical evaluation but also objective biological markers. Gingival crevicular fluid represents a reliable and non-invasive biomarker that reflects the biological response of peri-prosthetic soft tissues [14]. Evaluating the relationship between TGF-β2 levels in GCF and clinical healing indices may provide insight into the biological mechanisms of soft tissue maturation [13, 14].

A significant number of studies support the positive effects of laser therapy on healing oral soft tissue. These studies are mainly based on periodontal surgery, mucogingival procedures, and peri-implant soft tissue management, as well as experimental and animal studies [9, 11, 12, 15–21]. However, there is limited data on the impact of laser therapy on the maturation of the pontic site in patients planned for fixed partial denture rehabilitation. To date, there is a lack of well-designed clinical studies specifically investigating soft tissue healing at the pontic site in the context of fixed partial denture rehabilitation. In particular, the relationship between clinical healing outcomes and underlying biological responses, such as growth factor expression, remains insufficiently

explored in this specific prosthodontic setting. Therefore, the aim of this randomized placebo-controlled clinical trial was to evaluate the effect of laser-assisted therapy on soft tissue healing at the pontic site, using clinical parameters and biochemical analysis of gingival crevicular fluid, with particular emphasis on prosthodontically relevant outcomes. It was hypothesized that laser-assisted therapy would enhance soft tissue healing, reduce postoperative pain, and increase TGF-β2 levels compared with placebo.

Materials and methods

Study design

The study was conducted at the Faculty of Dentistry, Department of Prosthodontics, at Gaziantep University. The study protocol was approved by Gaziantep University Faculty of Medicine's Ethics Committee (decision number 2022/419), and all participants provided written informed consent prior to inclusion. The study was conducted in accordance with the principles of the Declaration of Helsinki.

A power analysis was performed using G*Power software (version 3.1.9.2; Heinrich Heine University, Düsseldorf, Germany). Based on a clinically relevant effect size of 0.45 derived from previous studies evaluating soft tissue healing outcomes, with a power of 80% and a two-sided significance level of 5%, the minimum required sample size was calculated as 45 participants [15–16]. Accordingly, a total of 45 patients were included in the study, with 15 patients allocated to each group.

Randomization and study groups

A total of 45 patients requiring single-tooth extraction followed by rehabilitation with a three-unit fixed partial denture were enrolled. All participants were systemically healthy non-smokers with good oral hygiene and no active periodontal disease. Patients with uncontrolled systemic conditions, pregnancy, a history of head and neck radiotherapy, or medications affecting wound healing were excluded.

Participants were randomly assigned to three groups ($n = 15$ per group): GRR laser (GRR), Nd: YAG laser (ND), and placebo (PG). Randomization was performed using a computer-generated sequence (SPSS v24.0, SPSS Inc., Chicago, IL, USA) by an independent researcher not involved in recruitment or treatment. Allocation concealment was ensured using sequentially numbered, sealed, opaque envelopes.

The study was designed as a single-blind trial. Participants were blinded to group allocation, while the operator could not be blinded due to the nature of the intervention. Outcome assessments were performed by an independent blinded examiner. The study was conducted in accordance with CONSORT guidelines, and the participant flow is presented in Fig. 1.

Treatment protocol

Laser applications were initiated immediately following tooth extraction and continued once daily for five consecutive days in all groups. All participants completed the study without loss to follow-up, ensuring the planned statistical power.

GRR laser group (GRR)

A Gallium–Aluminum–Arsenide diode laser (GRR Laser Medikal, Ankara, Turkey) was applied in non-contact mode for two minutes on the vestibular and two minutes on the lingual aspect (total irradiation time: 4 min per session). The probe was maintained at approximately 1 cm from the tissue surface to standardize application conditions.

The device combines an infrared laser (22 mW, 904 nm) and a red diode laser (10 mW, 650 nm), delivered simultaneously through a probe with a diameter of 0.6 cm (irradiation area ≈ 0.28 cm^2). According to the manufacturer, the device delivers 16 J/min; thus, the total energy per session was approximately 64 J, corresponding to an energy density of approximately 228 J/cm^2.

Nd: YAG laser group (ND)

A Nd: YAG laser (1064 nm; Fidelis Plus III, Fotona, Ljubljana, Slovenia) was used in pulsed mode according to the manufacturer's standard clinical settings. Irradiation was applied at two predefined points (vestibular and lingual aspects) using a standardized protocol.

The probe diameter was 0.9 cm, and the tip was positioned at a fixed distance of 1 cm from the mucosal surface. Laser irradiation was delivered with an output power of 0.25 W for 20 s per point, resulting in 5 J per point and an energy density of 8 J/cm^2. The total energy delivered per session was 10 J.

Placebo group (PG)

The placebo group followed the same procedural protocol as the Nd: YAG group; however, no laser energy was delivered.

Clinical evaluation parameters

The following clinical parameters were evaluated during follow-up:

- Wound dimensions: Mesiodistal width of the pontic site measured using a digital caliper.
- Postoperative pain: Assessed using a visual analog scale (VAS; 0–10).
- Soft tissue healing: Evaluated using the Landry, Turnbull and Howley Healing Index (HI) [17].

Clinical evaluations were performed at baseline (day of extraction), day 7, and day 14 by a blinded examiner.

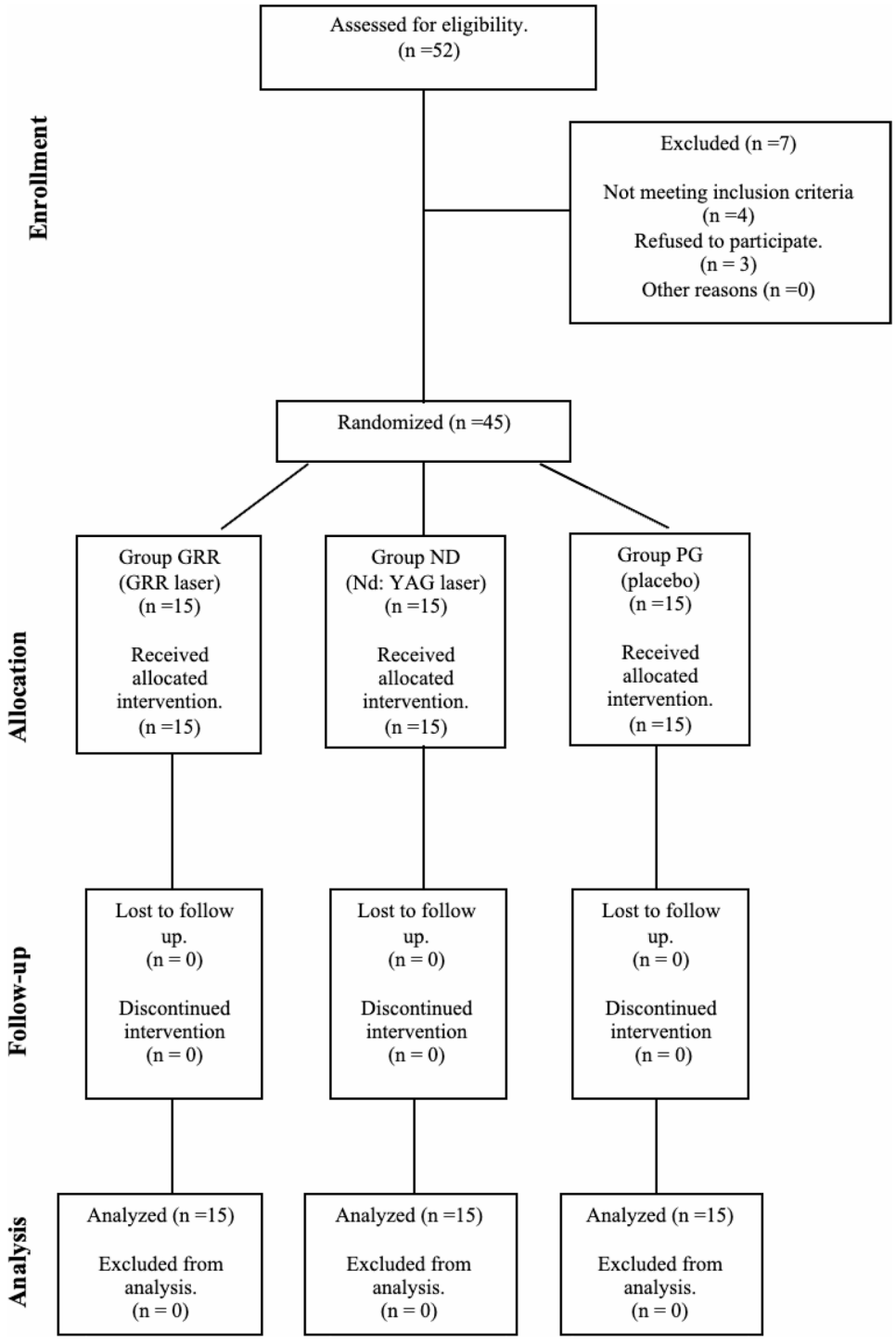


Fig. 1 CONSORT flow diagram

Wound dimensions were recorded immediately after extraction and at follow-up visits to assess dimensional changes during healing.

VAS scores were recorded daily from postoperative day 1 to day 7. Patients were instructed to avoid analgesic use unless absolutely necessary; no analgesic use was reported. VAS was not assessed at day 14, as all patients reported absence of pain.

Soft tissue healing was assessed using the HI, a validated clinical index evaluating gingival color, bleeding on palpation, wound margin integrity, granulation tissue formation, and suppuration. Healing was scored on an ordinal scale from 1 (very poor) to 5 (excellent).

Table 1 Evaluation of TGF-β2 (pg/mL) values within and among groups at baseline, day 7, and day 14. Intergroup and intragroup comparisons of TGF-β2 levels at baseline (preoperative), day 7, and day 14 are presented in Table 1

TGF pg/ Ml	GRR	ND	PG	p^1
	mean ± SD	mean ± SD	mean ± SD	
Preop	0.42 ± 0.05**A**	0.39 ± 0.04**A**	0.41 ± 0.05**A**	0.303
Day 7	1.38 ± 0.05**B***b*	0.89 ± 0.06**B***a*	0.88 ± 0.12**B***a*	**< 0.001**
Day 14	1.70 ± 0.06**C***c*	1.21 ± 0.14**C***b*	1.02 ± 0.10**C***a*	**< 0.001**
p^2	**< 0.001**	**< 0.001**	**< 0.001**	

Means followed by distinct capital letters in the same column, and distinct italic small letters in the same line are significantly different $p < 0.05$

[1]One way ANOVA

[2]Repeated measure ANOVA, Bold $p < 0.05$

Prior to the evaluations, the examiner was calibrated for both wound dimension measurements and clinical healing index scoring through repeated assessments under standardized clinical conditions. All measurements were performed by the same examiner using a standardized protocol to ensure consistency and minimize intra-examiner variability.

Gingival crevicular fluid sampling

GCF samples were collected to evaluate the biological response of peri-prosthetic soft tissues. At baseline, samples were obtained from the mesial and distal sulcus of the tooth scheduled for extraction. At days 7 and 14, GCF samples were collected from the sulcus of the adjacent abutment teeth closest to the pontic site. Due to the change in anatomical sampling site following extraction, baseline values were considered descriptive reference values rather than directly comparable longitudinal measurements. Sampling sites were isolated with cotton rolls and gently air-dried for 10 s. PerioPaper strips (PerioPaper, Oroflow, Amityville, NY) were inserted into the gingival sulcus for 30 s without inducing bleeding. Two samples were collected per site and immediately quantified using a Periotron (Oraflow Inc., Smithtown, NY, USA) device. Samples were then placed in Eppendorf tubes (Eppendorf tubes, Interlab, Istanbul, Turkey) and stored at − 80 °C until biochemical analysis.

Biochemical analysis of TGF-β2

All GCF samples were analyzed for TGF-β2 concentrations using a commercially available enzyme-linked immunosorbent assay (ELISA) kit (Elabscience, Europe), following the manufacturer's instructions. Standard curves were generated and used for quantitative analysis. The detection limit of the ELISA kit (E-EL-H1587, Elabscience) was 18.75 pg/mL, according to the manufacturer's specifications. Briefly, samples were diluted using the provided sample and standard dilution buffer and incubated for 24 h. The ELISA kit was equilibrated to room temperature for 30 min prior to use. Standards and samples (100 µL) were added to each well and incubated at 37 °C for 90 min. After removal of the liquid phase, biotin-labeled detection antibody (100 µL) was added and incubated at 37 °C for 60 min. Plates were washed three times with wash buffer, followed by addition of HRP conjugate (100 µL) and incubation at 37 °C for 30 min. After five washing cycles, substrate solution (90 µL) was added and incubated for 15 min at 37 °C. The reaction was stopped by adding stop solution (50 µL), and absorbance was measured at 450 nm using a microplate reader. Standard curves were generated and used for quantitative analysis.

Statistical analysis

The Shapiro–Wilk test was used to assess the normality of the data distribution. Parametric tests were applied for variables showing normal distribution. For TGF-β2 (pg/mL) values, repeated measures ANOVA (post hoc: Bonferroni) was used for intra-group comparisons across time points (baseline, day 7, and day 14). One-way ANOVA (post hoc: LSD) was used for intergroup comparisons. Time × group interactions were analyzed using two-way ANOVA. Mauchly's test of sphericity was used to assess the assumption of sphericity for repeated measures. Wound dimension data were analyzed using repeated measures ANOVA for intra-group comparisons and one-way ANOVA for intergroup comparisons. VAS scores were analyzed using non-parametric tests due to their ordinal nature. The Friedman test was used for intra-group comparisons, and the Kruskal–Wallis test was used for intergroup comparisons. The clinical healing index, representing ordinal data, was analyzed using non-parametric tests. The Friedman test was used for intra-group comparisons, and the Kruskal–Wallis test was used for intergroup comparisons. Descriptive statistics were presented as mean (standard deviation) for parametric data and median (interquartile range) for non-parametric data. All statistical analyses were performed using SPSS software (version 24.0, SPSS Inc., Chicago, IL, USA), with a significance level set at $p < 0.05$.

Results

TGF-β2 Levels in GCF

At baseline, no statistically significant differences were observed among the GRR, Nd: YAG, and placebo groups with respect to TGF-β2 levels ($p = 0.303$). However, statistically significant intergroup differences were detected at both day 7 and day 14 ($p < 0.001$). Post hoc analysis revealed that TGF-β2 levels at day 7 were significantly higher in the GRR laser group compared with the Nd: YAG and placebo groups, which showed comparable values ($p < 0.05$). At day 14, the lowest TGF-β2 levels were observed in the placebo group, followed by the Nd: YAG

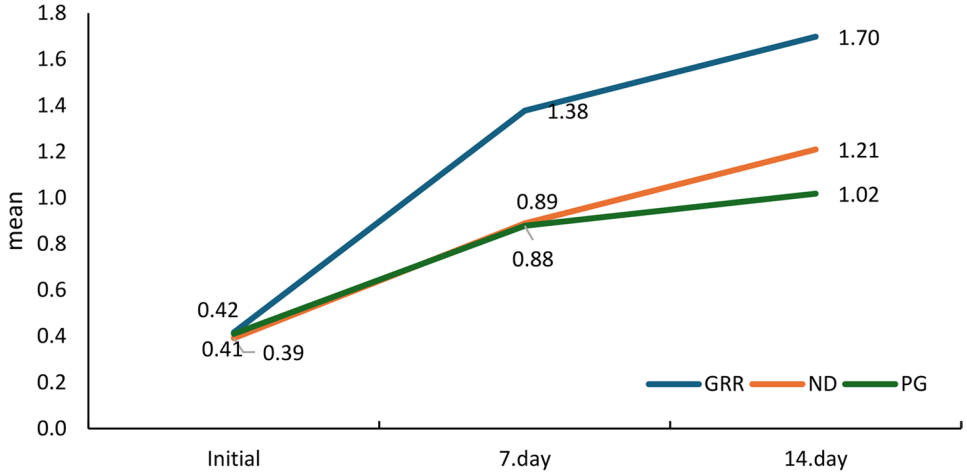


Fig. 2 Change in TGF pg/ml values over time in the GRR, ND and PG groups

Table 2 Evaluation of wound size parameters within and among groups at baseline, day 7, and day 14

Wound Dimensions (mm)	GRR mean ± SD	ND mean ± SD	PG mean ± SD	p^1
Preop	13.07 ± 0.88**A**	12.47 ± 1.36**A**	12.40 ± 1.12**A**	0.221
Day 7	5.07 ± 0.7**B***a*	7.27 ± 1.03**B***b*	7.27 ± 1.03**B***b*	**< 0.001**
Day 14	2.07 ± 0.88**C***a*	4.53 ± 0.64**C***b*	5.07 ± 0.7**C***b*	**< 0.001**
p^2	**< 0.001**	**< 0.001**	**< 0.001**	

Means followed by distinct capital letters in the same column, and distinct italic small letters in the same line are significantly different $p < 0.05$

[1]One way ANOVA

[2]Repeated measure ANOVA, Bold $p < 0.05$

group, whereas the highest levels were recorded in the GRR laser group ($p < 0.05$).

Intragroup comparisons demonstrated a statistically significant increase in TGF-β2 levels over time in all groups ($p < 0.001$). Pairwise analyses indicated that both day-7 and day-14 TGF-β2 levels were significantly higher than baseline values in the GRR, Nd: YAG, and placebo groups ($p < 0.05$).

Two-way ANOVA revealed a statistically significant main effect of time ($p < 0.001$), group ($p < 0.001$), and a significant group × time interaction ($p < 0.001$) on TGF-β2 levels. Post hoc analyses showed that the magnitude of TGF-β2 change from baseline to day 14 was significantly lower in the placebo group compared with the laser groups, and significantly lower in the Nd: YAG group compared with the GRR group ($p < 0.05$). Accordingly, the extent of TGF-β2 increase followed the order placebo < Nd: YAG < GRR, as illustrated in Fig. 2.

Pontic site wound dimensions

Wound size measurements (mm) demonstrated normal distribution and were analyzed using parametric tests. Intergroup and intragroup comparisons are summarized in Table 2.

No statistically significant differences in baseline wound size were observed among the three groups ($p = 0.221$). At day 7 and day 14, wound size values differed significantly among groups ($p < 0.001$). Post hoc analysis indicated that the Nd: YAG and placebo groups

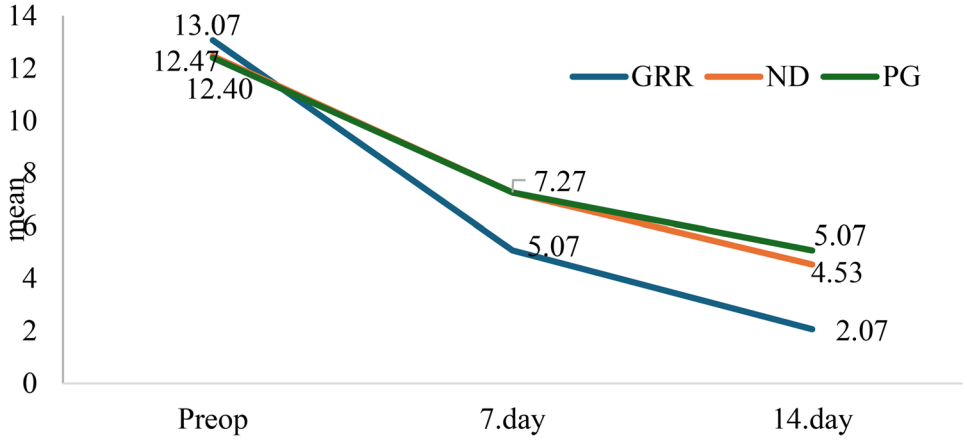


Fig. 3 Change in wound size (mm) over time in the GRR, ND and PG groups

Table 3 Evaluation of HI values within and among groups at baseline, day 7, and day 14

HI	GRR	ND	PG	p^1
	mean ± SD	mean ± SD	mean ± SD	
Preop	1.2 ± 0.41**A**	1.33 ± 0.49**A**	1.33 ± 0.49**A**	0.655
Day 7	4.33 ± 0.49**B***b*	2.53 ± 0.52**B***a*	2.6 ± 0.51**B***a*	**< 0.001**
Day 14	4.8 ± 0.41**C***b*	3.8 ± 0.56**C***a*	3.53 ± 0.52**C***a*	**< 0.001**
p^2	**< 0.001**	**< 0.001**	**< 0.001**	

Means followed by distinct capital letters in the same column, and distinct italic small letters in the same line are significantly different $p < 0.05$

[1]One way ANOVA

[2]Repeated measure ANOVA, Bold $p < 0.05$

Table 4 Evaluation of VAS scores within and among groups at postoperative days 1–7

VAS	GRR	ND	PG	p^1
	mean ± SD	mean ± SD	mean ± SD	
Day 1	6,27 ± 1,16**B***b*	6,8 ± 1,42**C***b*	5,73 ± 1,16**B***a*	**< 0,001**
Day 2	3,2 ± 1,7**B***a*	5,2 ± 1,21**BC***b*	4,87 ± 1,51**B***b*	**< 0,001**
Day 3	1,27 ± 0,8**A***a*	4,33 ± 1,29**BC***b*	3,93 ± 1,53**B***b*	**< 0,001**
Day 4	0,4 ± 0,63**A**	2,8 ± 1,52**AB**	2,73 ± 1,33**A**	0,655
Day 5	0,13 ± 0,35**A***a*	1,73 ± 0,88**A***b*	1,73 ± 1,16**A***b*	**< 0,001**
Day 7	0 ± 0**A***a*	1,13 ± 1,06**A***b*	1 ± 0,76**A***b*	**< 0,001**
p^2	**< 0,001**	**< 0,001**	**< 0,001**	

Means followed by distinct capital letters in the same column, and distinct italic small letters in the same line are significantly different $p < 0.05$

[1]Kruskal Wallis

[2]Friedman test, Bold $p < 0.05$

exhibited comparable wound sizes at day 7, whereas the GRR group showed significantly greater reduction in wound size compared with the other groups ($p < 0.05$). At day 14, the smallest wound dimensions were observed in the GRR group, followed by the Nd: YAG group, while the largest values were recorded in the placebo group ($p < 0.001$).

Within-group analyses revealed a statistically significant reduction in wound size from baseline to day 7 and day 14 in all groups ($p < 0.001$). Two-way ANOVA demonstrated significant effects of time ($p < 0.001$), group ($p < 0.001$), and group × time interaction ($p < 0.001$). The magnitude of wound size reduction from baseline to day 14 was lowest in the Nd: YAG group, followed by the placebo group, and highest in the GRR group (Nd: YAG < placebo < GRR), as shown in Fig. 3.

Clinical healing index (HI)

HI scores showed normal distribution and were evaluated using parametric analyses. Intergroup and intragroup comparisons are presented in Table 3.

Baseline HI scores did not differ significantly among the groups ($p = 0.655$). At both day 7 and day 14, statistically significant intergroup differences were observed ($p < 0.001$). Post hoc analysis revealed that HI scores in the GRR group were significantly higher than those in the Nd: YAG and placebo groups at both time points, whereas no significant difference was observed between the Nd: YAG and placebo groups ($p < 0.001$).

Intragroup comparisons showed a significant increase in HI scores over time in all groups ($p < 0.001$). Two-way ANOVA demonstrated significant main effects of time ($p < 0.001$), group ($p < 0.001$), and a significant group × time interaction ($p < 0.001$). The magnitude of HI improvement from baseline to day 14 was significantly greater in the GRR group compared with the Nd: YAG and placebo groups, which showed comparable changes (Nd: YAG = placebo < GRR), as illustrated in Fig. 4.

Pain Assessment (VAS Scores).

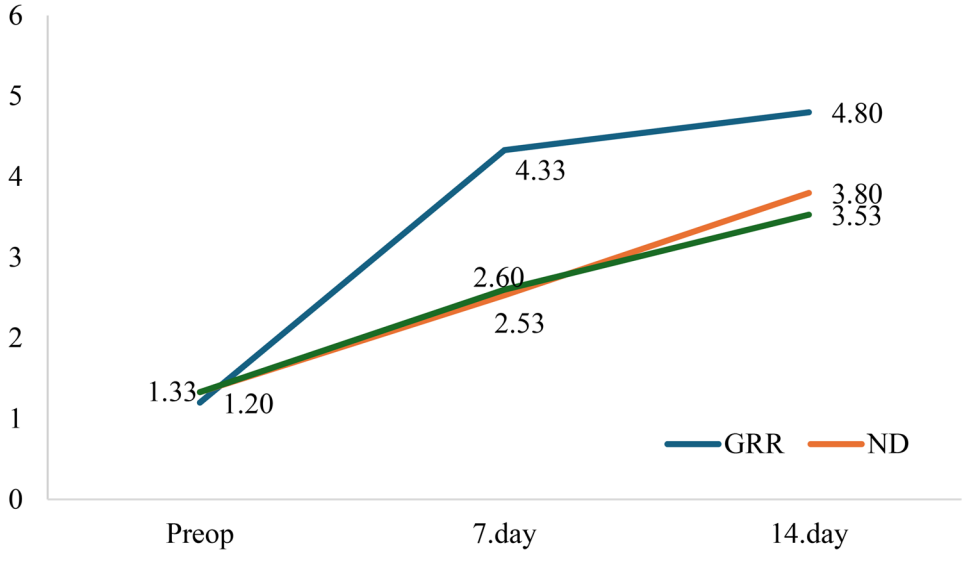


Fig. 4 Change in HI values over time in the GRR, ND and PG groups

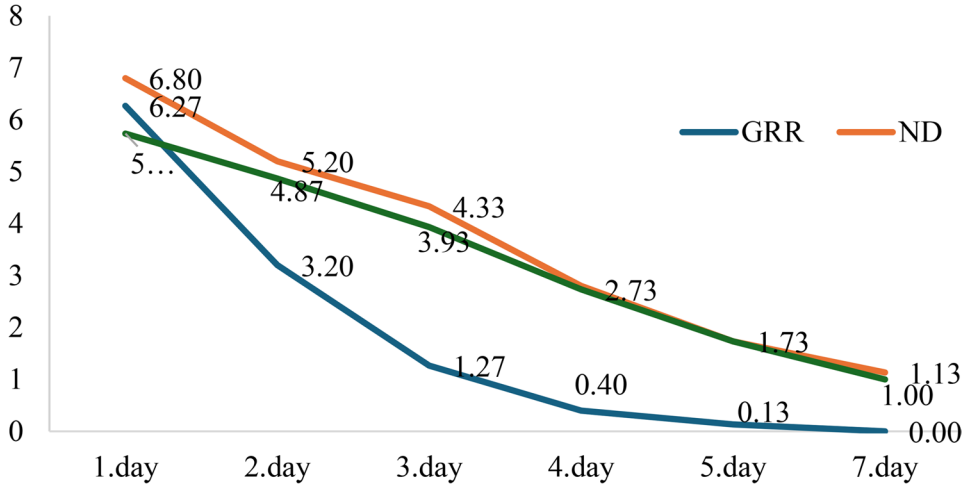


Fig. 5 Change in VAS scores over time in the GRR, ND and PG groups

Intergroup and intragroup comparisons from day 1 to day 7 are summarized in Table 4.

Statistically significant differences in VAS scores were observed among the groups at all time points except day 4 ($p < 0.001$). On day 1, the placebo group demonstrated significantly lower VAS scores compared with the GRR and Nd: YAG groups ($p < 0.001$). On days 2, 3, 5, and 7, VAS scores in the GRR group were significantly lower than those in the Nd: YAG and placebo groups, which showed comparable values ($p < 0.001$).

Within-group analyses indicated that a statistically significant reduction in VAS scores occurred after day 2 in the GRR group, whereas significant reductions were observed after day 4 in the Nd: YAG and placebo groups ($p < 0.05$). Temporal changes in pain scores are illustrated in Fig. 5.

Discussion

This randomized, placebo-controlled clinical trial investigated whether laser-assisted therapy could improve soft tissue healing at the pontic site prior to fixed partial denture rehabilitation. This was assessed using both clinical healing outcomes and a biological marker obtained from GCF. The study's primary finding was that laser therapy produced measurable improvements in pontic site healing dynamics compared to the placebo group, with the most noticeable effects observed in the GRR laser group. Consequently, the null hypothesis that laser-assisted therapy would not result in significant differences in pontic site soft tissue healing or biological healing markers compared with the placebo group was rejected.

LLLT or photobiomodulation (PBM) is generally defined as a non-thermal adjunctive therapy that accelerates soft tissue repair. It facilitates the transition of the early-stage wound area from an inflammatory to a regenerative state, alleviating pain in the process. The literature emphasises that low-level laser therapy LLLT primarily targets cellular mitochondria. Cytochrome c oxidase is considered a key photoreceptor and its effects include increased mitochondrial activity and adenosine triphosphate (ATP) production, which support cellular metabolism and early tissue repair [18]. LLLT is similarly defined as a method that regulates critical cellular pathways via signals such as ATP, calcium and reactive oxygen species, and that promotes collagen formation, granulation tissue and revascularisation. These processes directly contribute to faster epithelialisation and improved soft tissue quality [19]. When examining the evidence base across different studies, the results are heterogeneous; however, they generally support a short-term benefit depending on wavelength, energy density, timing, and the tissue being treated. A systematic review examining healing after tooth extraction (16 studies; 8 clinical/8 animal) concluded that laser treatment accelerated healing, but noted that several studies reported no improvement and that benefits appeared limited to specific laser types and settings, emphasising protocol sensitivity [20]. The importance of the LLLT protocol is also evident in the studies conducted [15, 21, 22]. In one study, a 980 nm diode laser was applied to the wound area. While the extraction sockets showed significantly smaller non-epithelialised surfaces on days 3 and 7 (improvement in soft tissue closure), pain reduction remained limited (the difference was mainly observed on day 3) [15]. In contrast, a study involving a large patient population undergoing routine tooth extraction followed by PBM therapy reported that pain and inflammation decreased over time in both the treatment and sham groups, with no clear evidence of an additional improvement advantage over the sham group. This may again be due to differences in the selected cases or protocols [23]. A randomized clinical trial involving more invasive third molar surgery

demonstrated that the collagen and laser combination was superior in terms of short-term recovery, with differences between groups decreasing by day 7. This is consistent with the idea that LLLT has its most clinically significant effect during the early inflammatory phase [24]. The differences observed between the Nd: YAG and GRR laser groups in the present study may be explained by variations in wavelength characteristics and energy delivery patterns, as well as how these interact with the biology of soft tissue healing. Nd: YAG lasers, operating in the near-infrared spectrum, GRR laser system used in this study combines infrared and red wavelengths, which may have different biological effects at various depths of tissue. Red-spectrum wavelengths are known to interact preferentially with the superficial layers of the epithelium and fibroblasts, promoting keratinocyte migration, fibroblast proliferation and collagen synthesis [25]. Infrared wavelengths, on the other hand, may support deeper vascular and inflammatory modulation. The combined delivery of these wavelengths may therefore provide a broader photobiomodulatory stimulus across multiple phases of wound healing, resulting in enhanced growth factor expression and connective tissue maturation. In addition to wavelength-related differences, variations in application protocols may have significantly influenced the outcomes observed between the laser groups. Parameters such as energy density, irradiation time, mode of energy delivery, and total energy applied per session are critical determinants of photobiomodulation effects. Nd: YAG lasers are typically associated with deeper tissue penetration and may involve different energy delivery dynamics, whereas the GRR system used in this study applies low-power combined wavelengths over a longer exposure period, favoring cumulative photobiomodulatory stimulation. These protocol-related differences may influence the balance between anti-inflammatory effects and regenerative responses, and could partially explain the differences observed between the laser groups. This highlights the importance of protocol standardization when interpreting and comparing laser-assisted therapies. This hypothesis is supported by the significantly higher TGF-β2 levels observed in the GRR group, which were accompanied by superior clinical healing indices and greater wound size reduction.

The reason for selecting TGF-β2 as a biological marker in the present study was its well-established role in soft tissue wound healing and extracellular matrix remodelling. Studies have demonstrated that TGF-β signalling is particularly active during the early and intermediate phases of wound healing. During this time, it contributes to the formation of granulation tissue and the maturation of connective tissue [26–28]. A comprehensive review of TGF-β isoforms has emphasised that the differential expression patterns of TGF-β1, TGF-β2 and TGF-β3 may reflect the specific phases and biological characteristics of wound repair, particularly in mucosal tissues [29]. Gingival crevicular fluid is a widely used non-invasive medium for monitoring local growth factor activity during healing and regeneration. Previous studies have reported that changes in TGF-β levels in this fluid correlate with clinical healing and tissue remodelling processes, thus supporting its validity as a biomarker of the local biological response. Furthermore, literature focusing on growth factors highlights TGF-β isoforms, including TGF-β2, as key mediators linking cellular activity to clinical wound healing outcomes [30–33]. Therefore, evaluating TGF-β2 levels in gingival crevicular fluid was considered an appropriate way to reflect changes in the healing of soft tissue at the pontic site that are biologically relevant, and to complement clinical healing indices with an objective biochemical parameter. From a clinical perspective, the observed increase in TGF-β2 levels may reflect enhanced connective tissue organization and more rapid transition from the inflammatory to the proliferative phase of healing. This may contribute to improved soft tissue stability, reduced risk of prolonged inflammation, and more favorable tissue conditions prior to prosthetic rehabilitation. However, it should be noted that the present study did not directly evaluate long-term clinical outcomes related to prosthetic performance, and therefore these interpretations should be considered within the context of early healing responses.

The rapid and successful healing of soft tissue at the pontic site is critical to achieving both biological stability and prosthetic success in fixed partial denture rehabilitation. Following tooth extraction, dimensional changes to the alveolar ridge and the surrounding soft tissues are frequently observed. These changes can lead to ridge collapse, loss of soft tissue contour and compromised pontic site morphology. If the pontic recipient site does not heal in a timely and organised manner, these post-extraction changes may adversely affect the design, aesthetics, phonetics, cleanability and long-term prognosis of fixed dental prostheses. Literature emphasises that achieving a well-healed and stable soft tissue–prosthetic interface is essential for predictable aesthetic and biological outcomes, particularly at pontic sites where tissue contact or controlled pressure is required [34–37]. Inadequate or delayed healing of the pontic region may require prolonged provisionalisation, repeated modification of the pontic contours or further soft tissue augmentation procedures. This increases the duration and complexity of the treatment. Clinical reports and reviews on pontic site management have consistently demonstrated that stable soft tissue volume and quality are essential for achieving natural-looking pontic contours and minimising biological complications, such as inflammation, ulceration, food impaction and patient discomfort [4–39]. In this respect, adjunctive methods that are biologically safe and capable

of accelerating soft tissue healing at the pontic site are of considerable clinical importance. The findings of the present study suggest that laser-assisted therapy, particularly the GRR protocol, was associated with improved early soft tissue healing parameters. These findings may contribute to more favorable soft tissue conditions prior to prosthetic rehabilitation.

When considering the findings, it is important to acknowledge the limitations of the present study. Firstly, the follow-up period was limited to 14 days, reflecting the early and intermediate phases of soft tissue healing at the pontic site. While this timeframe is clinically relevant for evaluating initial wound closure, tissue maturation and preparation for subsequent prosthetic procedures, it does not allow conclusions regarding long-term soft tissue stability following definitive fixed partial denture placement. The 14-day follow-up period was specifically designed to evaluate early wound healing phases, where photobiomodulation effects are most pronounced. Secondly, although GCF analysis is a reliable and non-invasive method for evaluating local biological responses, the change in anatomical sampling site following tooth extraction may introduce biological variability and limit direct longitudinal comparisons between baseline and follow-up measurements. Therefore, changes in TGF-β2 levels should be interpreted with caution, as they may reflect both healing-related biological responses and, to some extent, differences between sampling locations. Following extraction, sampling from the original site is not feasible due to the absence of a gingival sulcus; therefore, adjacent teeth within the same local microenvironment were selected as representative sites. Although this approach provides standardized and clinically feasible sampling conditions, it may not fully represent the molecular activity directly beneath the pontic contact area. Further studies incorporating extended follow-up periods, additional biomarkers and prosthodontic performance metrics would be valuable in clarifying the clinical implications of laser-assisted therapy in pontic site management.

Conclusion

Within the limitations of this randomized, placebo-controlled clinical trial, laser-assisted therapy was associated with improved early soft tissue healing parameters at the pontic site prior to fixed partial denture rehabilitation. Among the evaluated protocols, the GRR laser was associated with more favorable clinical and biological outcomes, as reflected by improved healing index scores, greater wound size reduction, and higher TGF-β2 levels in GCF compared with the Nd: YAG laser and placebo groups. However, the present study evaluated only short-term healing outcomes during the early phases of soft tissue repair, and these findings should not be interpreted as evidence of long-term prosthetic success or long-term soft tissue stability. In addition, while the biological rationale is supported by existing literature, much of the available evidence is derived from general oral wound healing rather than pontic-specific clinical conditions. Therefore, the findings should be interpreted within this context. Further well-designed clinical studies with longer follow-up periods and prosthodontic outcome measures are needed to better understand the clinical implications of laser-assisted therapy in pontic site management.

Abbreviations

ANOVA	Analysis of Variance
ATP	Adenosine Triphosphate
ELISA	Enzyme-Linked Immunosorbent Assay
FPD	Fixed Partial Denture
GCF	Gingival Crevicular Fluid
GRR	GRR Laser Group
HI	Landry, Turnbull and Howley Healing Index
LLLT	Low-Level Laser Therapy
ND	Nd: YAG Laser Group
Nd	YAG: Neodymium-Doped Yttrium Aluminum Garnet
PBM	Photobiomodulation
PG	Placebo Group
SPSS	Statistical Package for the Social Sciences
TGF-β2	Transforming Growth Factor Beta 2
VAS	Visual Analog Scale

Supplementary Information

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Supplementary Material 1.

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Not Applicable.

Authors' contributions

I.K., O.P.O, N.D conceived and designed the study. L.T and O.P.O performed the clinical procedures and data collection. S.T. performed the biochemical analysis. I.K. drafted the manuscript. IK and MD made adjustments in line with the revision. Authors critically revised the manuscript and approved the final version.

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Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

The study protocol was approved by the Ethics Committee of Gaziantep University Faculty of Medicine (Decision number: 2022/419) and was performed in accordance with the principles of the Declaration of Helsinki. Written informed consent was obtained from all participants prior to enrollment in the study.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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